Block Matrices

A large matrix often has structure that its entries hide. Split it into blocks, smaller matrices arranged in a grid, and the structure shows. A direct sum of subspaces becomes a block-diagonal matrix. A subspace that an operator maps into itself becomes a block of zeros in the corner. A system of equations that splits into two groups becomes a two-by-two system whose "entries" are matrices.

This chapter makes block arithmetic rigorous and then puts it to work. Multiplying block by block, eliminating a whole block at once, and taking the Schur complement $\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$ turn statements about big matrices into statements about small ones: ranks, determinants, and inverses. The Kronecker product rewrites matrix equations such as $\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{B} = \mathbf{C}$ as ordinary linear systems. The chapter ends with a question that looks like it needs heavy machinery but does not: which matrices can be written as a commutator $\mathbf{X}\mathbf{Y} - \mathbf{Y}\mathbf{X}$?

What you need. From Chapter 0, matrix multiplication (Definition 0.9.9) and polynomials without zero divisors (Corollary 0.8.4). From Chapter 1, subspaces, spans, and bases with the dimension formula for a sum (Theorem 1.3.1, Theorem 1.6.5, Theorem 1.6.3, Theorem 1.6.7, Proposition 1.7.2, Theorem 1.7.3), the unique representation of a vector in a basis (Theorem 1.5.1, Theorem 1.4.3), and the direct sum criteria (Theorem 1.7.5), which is what §01 reads off a block decomposition; §05 also uses the standard bases and the trace-free matrices as examples (Example (Standard Bases), Example (Trace-Free Matrices), Theorem 1.1.9). From Chapter 3, rank (Definition 3.5.4). From Chapter 2, the matrix of a linear map and change of basis (Definition 4.1.1, Theorem 4.2.3), quotient spaces (Definition 4.4.3), the rank inequalities (Theorem 4.3.5), and the trace (Definition 4.5.2). From Chapter 7, determinants of block triangular matrices and the characteristic polynomial (Theorem 7.8.4, Theorem 7.10.2).

8.-1.1 Roadmap

- **Partitions and block multiplication.** Multiplying matrices block by block, and what block-diagonal and block-triangular shapes say about a linear map.
- **Rank and block matrices.** Rank inequalities proved by clearing blocks, including Frobenius's inequality.
- **Schur complements.** The Schur complement, the determinant, rank, and inverse of a block matrix, the Woodbury formula, and determinants with commuting blocks over any field.
- **The Kronecker product.** A product of matrices of different sizes, and matrix equations turned into linear systems.
- **Commutators and trace-zero matrices.** Every trace-zero matrix over a field of characteristic zero is a commutator.

8.-1.2 Named moves

- **Partition to match the structure.** Choose the blocks so that the zeros you know about sit in whole blocks.
- **Eliminate a block.** Multiply by an invertible block triangular matrix to clear a block, just as a row operation clears an entry.
- **Pass to the Schur complement.** Reduce a question about $\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$ with $\mathbf{A}$ invertible to a question about $\mathbf{A}$ and $\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$.
- **Vectorize.** Turn a linear equation in an unknown matrix $\mathbf{X}$ into a linear system in the

entries of $\mathbf{X}$.

- **Adjoin an indeterminate.** To remove an invertibility hypothesis, replace $\mathbf{D}$ by $\mathbf{D} + x\mathbf{I}$, work over $F[x]$, and set $x = 0$ at the end.